JEE Mains
9th April, 2019 Evening

Physics

1. In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28}\ m^{-3}$ and mean free time is 25 *f*s (femto second), it's approximate resistivity is:

$\left(m_e = 9.1 \times 10^{-31}\ kg\right)$

A. $10^{-7}\ \Omega m$ B. $10^{-6}\ \Omega m$

C. $10^{-8}\ \Omega m$ D. $10^{-5}\ \Omega m$

Ans. C

Sol: Resistivity is given by

$\rho = \frac{m_e}{\tau e^2 \eta}$ where τ = mean free time, η = conduction electrons per unit volume

$$= \frac{9.1 \times 10^{-31}}{25 \times 10^{-15} \times \left(1.6 \times 10^{-19}\right)^2 \times 8.5 \times 10^{28}}$$

$= 1.67 \times 10^{-8}\ \Omega m$

2. The specific heats, C_P and C_V of a gas of diatomic molecules, A, are given (in units of J mol^{-1} K^{-1}) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and 21. If they are treated as ideal gases, then:

A. Both A and B have a vibrational mode each.
B. A has a vibrational mode but B has none.
C. A has one vibrational mode and B has two.

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D. A is rigid but B has a vibrational mode.

Ans. B

Sol:

Given cases,

For gas A, $C_p = 29$, $C_v = 22$

For gas B, $C_p = 30$, $C_v = 21$

For A :

$$\frac{C_P}{C_V} = \frac{29}{22}$$

$$1 + \frac{2}{f} = \frac{29}{22}$$

$$\frac{2}{f} = \frac{7}{22}$$

$$f = \frac{44}{7} = 6.6$$

3 translation, 2 rotation, remaining vibration

For B :

$$1 + \frac{2}{f} = \frac{30}{21}$$

$$\frac{2}{f} = \frac{9}{21}$$

$$f = \frac{42}{9}$$

No vibrational

3. A wooden block floating in a bucket of water has $\frac{4}{5}$ of its volume submerged. When certain amount of an oil is poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is:

A. 0.8 B. 0.7

C. 0.6 D. 0.5

Ans. C

Sol:

For a floating body,

upthrust = weight of the part of object, i.e., submerged in the fluid.

For case 1

$$mg = F_3$$

$$mg = m'g$$

$$m = \frac{4}{5}Vd_w$$

For case 2

$$mg = F_{bw} + F_{bo}$$

$$m = \frac{V}{2}d_w + \frac{V}{2}d_0$$

By equation 1 and 2

$$\frac{4}{5}Vdw = \frac{V}{2}d_w + \frac{V}{2}d_0$$

$$\frac{d_0}{d_w} = \frac{6}{10} = 0.6$$

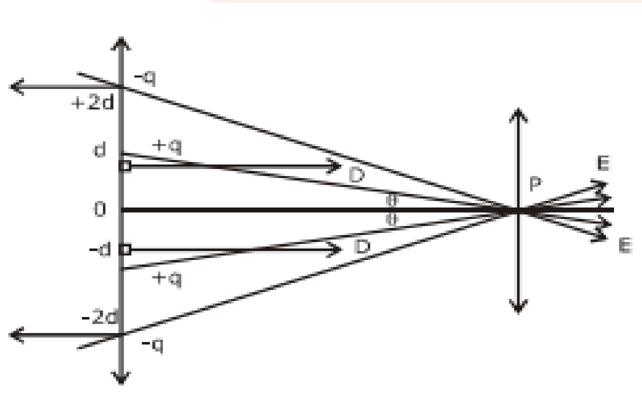
4. Four point charges –q, +q, +q and –q are placed on *y*-axis at *y* = –2d, *y* = –d, *y* = +d and *y* = +2d, respectively. The magnitude of the electric field E at a point on the *x*-axis at *x* = D, with D>>d, will behave as:

A. $E \propto \frac{1}{D^2}$ B. $E \propto \frac{1}{D^3}$

C. $E \propto \frac{1}{D^4}$ D. $E \propto \frac{1}{D}$

Ans. C.

Sol:



We know that,

$$E_P = 2E_1 \cos\theta_1 - 2E_1 \cos\theta_2$$

$$= \frac{2Kq}{d^2 + D^2} \times \frac{D}{\left(d^2 + D^2\right)^{3/2}} - \frac{2Kq}{(2d)^2 + D^2} \times \frac{D}{\left[(2d)^2 + D^2\right]^{3/2}}$$

$$= 2K\,qD\left[\left(d^2 + D^2\right)^{3/2} - \left(4d^2 + D^2\right)^{-3/2}\right] \qquad d << D$$

$$= \frac{2\,KqD}{D^2}\left(1 - \frac{3}{2}\frac{d^2}{D^2} - 1 - \frac{3 \times 4d^2}{2D^2}\right)$$

$$= \frac{2\,KqD}{D^2}\left(\frac{2D^2 - 3d^2}{2D^2} - \frac{2D^2 - 12d^2}{2D^2}\right)$$

$$= \frac{2\,KqD}{D^2}\left(\frac{2D^2 - 3d^2 - 2D^2 + 12d^2}{2D^2}\right)$$

$$= \frac{\cancel{2}\,KqD}{D^2}\left(\frac{\cancel{2D^2} - 3d^2 - \cancel{2D^2} + 12d^2}{\cancel{2}D^2}\right)$$

$$= \frac{9\,Kqd^2}{D^4}$$

5. 50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1 m^2 surface area will be close to ($c = 3 \times 10^8$ m/s):

A. 20×10^{-8} N　　B. 15×10^{-8} N

C. 35×10^{-8} N　　D. 10×10^{-8} N

Ans. A

Sol:

Radiation pressure or momentum imparted per second per unit area when light falls is

$$p = \begin{cases} \dfrac{2I}{c}; & \text{for reflection of radiation} \\ \dfrac{I}{c}; & \text{for absorption of radiation} \end{cases}$$

where, I is the intensity of the light.

The force exerted on 1 m^2 surface area will be

$$F = \frac{25}{100}\left(\frac{2I}{c}\right) + \frac{75}{100}\left(\frac{I}{c}\right)$$

$$= \frac{225}{100}\left(\frac{I}{c}\right)$$

$$= \frac{125}{100} \times \frac{50}{30 \times 10^8}$$

$$= 20.8 \times 10^{-8} \text{ N}$$

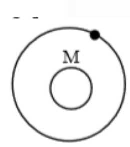
6. A test particle is moving in a circular orbit in the gravitational field produced by a mass density $\rho(r) = \frac{K}{r^2}$.
Identify the correct relation between the radius R of the particle's orbit and its period T:

A. T^2/R^3 is a constant B. T/R is a constant

C. TR is a constant D. T/R^2 is a constant

Ans. B

Sol:



Centripetal Force = Gravitational Field

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$dm = \left(4\pi r^2 dr\right)\rho$$

$$\int dm = \int 4\pi r^2 dr.\frac{K}{r^2}$$

$$m = 4\pi\ Kr$$

$$V = \sqrt{\frac{G4\pi Kr}{r}}$$

$$V = \sqrt{4\pi KG}$$

$$T = \frac{2\pi R}{V}$$

$$\frac{T}{R} \to \text{const.}$$

7. Two cars A and B are moving away from each other in opposite directions. Both the cars are moving with a speed of 20 ms^{-1} with respect to the ground. If an observer in car A detects a frequency 2000 Hz of the sound coming from car B, what is the natural frequency of the sound source in car B?(speed of sound in air = 340 ms^{-1})

A. 2150 Hz　　B. 2060 Hz

C. 2300 Hz　　D. 2250 Hz

Ans. D

Sol:

Source and observer both are moving away from each other. So, by Doppler's effect, observed frequency is given by

$$n' = n\left(\frac{v + v_0}{v - v_s}\right)$$

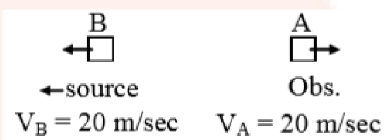
where, v = speed of sound = 340 ms^{-1}

v_0 = speed of observer = -20 ms^{-1}

v_s = speed of source = -20 ms^{-1}

n = true frequency

and n = apparent frequency = 2000 Hz



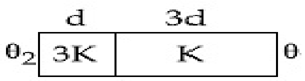
$$n' = n\left(\frac{v - v_0}{v + v_s}\right)$$

$$2000 = n\left(\frac{340 - 20}{340 + 20}\right)$$

$$2000 = n\left(\frac{320}{360}\right)$$

$$n = 2000 \times \frac{9}{8} = 2250 \text{ Hz}$$

8. Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are ‘θ_2’ and ‘θ_1 respectively. ($\theta_2 > \theta_1$). The temperature at the interface is:



A. $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$

B. $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

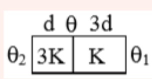
C. $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

D. $\frac{\theta_2 + \theta_1}{2}$

Ans. B

Sol:

Let interface temperature in steady state conduction is θ, then assuming no heat loss through sides;



Heat current will be same in both

$$H_1 = H_2$$

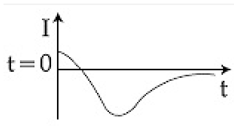
$$\frac{3KA}{d}\left(\theta_2 - \theta\right) = \frac{KA}{3d}\left(\theta - \theta_1\right)$$

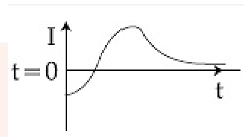
$$9\theta_2 - 9\theta = \theta - \theta_1$$

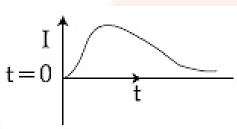
$$\theta = \frac{\theta_1 + 9\theta_2}{10}$$

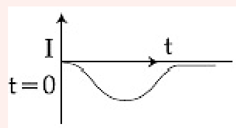
$$= \frac{\theta_1}{10} + \frac{9\theta_2}{10}$$

9. A very long solenoid of radius R is carrying current $I(t) = kte^{\alpha t}$ ($k>0$), as a function of time ($t \geq 0$). Counter clockwise current is taken to be positive. A circular conducting coil of radius 2R is placed in the equatorial plane of the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by:

A. 

B. 

C. 

D. 

Ans. B

Sol:

Magnetic flux associated with the outer coil is

$$\phi = \left(\mu_0 nKte^{-\alpha t}\right)4\pi R^2$$

Induced emf,

$$e = \frac{-d\phi}{dt} = -ce^{-\alpha t}\left(1-\alpha t\right)$$

$$i_{induced} = \frac{-ce^{-\alpha t}\left(1-\alpha t\right)}{R}$$

at $t = 0$

$i_{ind} \Rightarrow +v_c$

10. Two coils ‘P’ and ‘Q’ are separated by some distance. When a current of 3 A flows through oil ‘P’, a magnetic flux of 10^{-3} Wb passes through ‘Q’. No current is passed through

‘Q’. When no current passes through ‘P’ and a current of 2 A passes through ‘Q’, the flux through ‘P’ is:

A. 3.67×10^{-3} Wb B. 6.67×10^{-3} Wb

C. 3.67×10^{-4} Wb D. 6.67×10^{-4} Wb

Ans. D

Sol:

Flux is given by,

$$\phi = BA$$

$$\phi_P = \frac{\mu_0 i_1 R^2}{2\left(R^2 + x^2\right)^{3/2}} \pi r^2 = 10^{-3}$$

$$\phi_Q = \frac{\mu_0 i_2 r^2}{2\left(r^2 + x^2\right)^{3/2}} \pi R^2$$

$$\frac{\phi_P}{\phi_Q} = \frac{\dfrac{\mu_0 i_1 R^2}{2\left(R^2 + x^2\right)^{3/2}} \pi r^2}{\dfrac{\mu_0 i_2 r^2}{2\left(r^2 + x^2\right)^{3/2}} \pi R^2}$$

$$\frac{\phi_P}{\phi_Q} = \frac{\dfrac{\mu_0 i_1 R^2}{\cancel{2}\left(R^2 + x^2\right)^{3/2}} \pi r^2}{\dfrac{\mu_0 i_2 r^2}{\cancel{2}\left(r^2 + x^2\right)^{3/2}} \pi R^2}$$

$$\frac{\phi_P}{\phi_Q} = \frac{i_2}{i_1}\left(\frac{R^2 + x^2}{r^2 + x^2}\right)^{3/2}$$

$$\phi_P = \frac{2 \times 10^{-3}}{3} = 6.67 \times 10^{-4}$$

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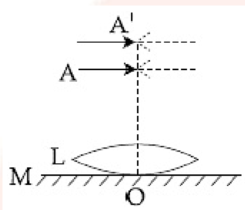
11. A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When a pin is placed at A, such that OA = 18 cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_l is put between the lens and the mirror, the pin has to be moved to A', such that OA' = 27 cm, to get its inverted real image at A' itself. The value of μ_l will be:



A. $\sqrt{3}$

B. $\sqrt{2}$

C. $\frac{3}{2}$

D. $\frac{4}{3}$

Ans. D

Sol:

For convex lens by lens maker's formula, we have

$$\frac{1}{f} = (\mu - 1)\left(\frac{2}{R}\right) \Rightarrow \frac{1}{18} = 0.5 \times \frac{2}{R} \Rightarrow R = 18\,\text{cm}$$

and for plane-convex lens by lens maker's formula, we have

$$\frac{1}{f} = (\mu_l - 1)\left(\frac{-1}{R}\right) \Rightarrow -\frac{1}{54} = (\mu_l - 1)\left(\frac{-1}{18}\right)$$

$$\Rightarrow \mu_l = 1 + \frac{1}{3} = \frac{4}{3}$$

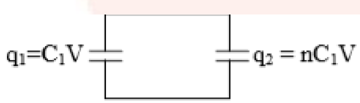
12. The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a batter of voltage, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is:

A. $\frac{(n+1)V}{(K+n)}$ B. V

C. $\frac{nV}{K+n}$ D. $\frac{V}{K+n}$

Ans. A

Sol: When battery is removed and a dielectric slab is placed between two plates of first capacitor, then charge on the system remains same. Now, equivalent capacitance after insertion of dielectric is



$q = CV + nCV$
$= (n+1)CV$
Due to insertion of dielectric
$q_1 = KCV_C$

$q_2 = nCV_C$

$V_C = \frac{q_{total}}{C_{eff}}$

$= \frac{(n+1)CV}{KC + nC} = \frac{(n+1)}{K+n}V$

13. A particle 'P' is formed due to completely inelastic collision of particles 'x' and 'y' having de-Broglie wavelengths ' λ_x' and 'λ_y' respectively. If x and y were moving in opposite directions, then the de-Broglie wavelength of 'P' is:

A. $\lambda_x + \lambda_y$

B. $\frac{\lambda_x \lambda_y}{|\lambda_x - \lambda_y|}$

C. $\lambda_x - \lambda_y$

D. $\frac{\lambda_x \lambda_y}{\lambda_x + \lambda_y}$

Ans. B

Sol:

We have, de-Broglie wavelengths associated with particles are

$\Rightarrow P_x = \frac{h}{\lambda_x}$ and $P \frac{h}{\lambda_y}$

The law of momentum conservation

$P_x + P_y = P$

$\frac{h}{\lambda_x} - \frac{h}{\lambda_y} = \frac{h}{\lambda} \Rightarrow \lambda = \frac{\lambda_x \lambda_y}{|\lambda_x - \lambda_y|}$

14. A wedge of mass M = 4m lies on a frictionless plane. A particle of mass m approaches the wedge with speed υ.

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There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by:

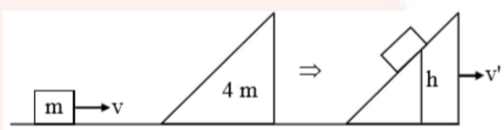
A. $\frac{\upsilon^2}{2g}$ B. $\frac{2\,\upsilon^2}{5g}$

C. $\frac{2\,\upsilon^2}{7g}$ D. $\frac{\upsilon^2}{g}$

Ans. B

Sol:

Since, the ground is frictionless, so when the particle will collide and climb over the wedge, then the wedge will also move. Thus, by using conservation laws for momentum and energy, maximum height climbed by the particle can be calculated.



conservation of momentum

$mv = 5\,mv'$

$v' = \frac{v}{5}$

By applying conservations of mechanical energy, we have

Initial energy of the system = Final energy of the system.

$$\frac{1}{2}mv^2 = \frac{1}{2}5\,mv'^2 + mgh$$

$$\frac{1}{2}v^2 = \frac{5}{2}\left(\frac{v}{5}\right)^2 + gh$$

$$\frac{v^2}{2} - \frac{v^2}{10} = gh$$
$$h = \frac{2v^2}{5g}$$

15. A string 2.0 m long and fixed at its ends is driven by a 240 Hz vibrator. The string vibrates in its third harmonic mode. The speed of the wave and its fundamental frequency is:

A. 320 m/s, 120 Hz
B. 180 m/s, 80 Hz
C. 320 m/s, 80 Hz
D. 180 m/s, 120 Hz

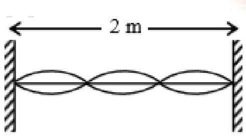
Ans. C

Sol:

Frequency of vibration of a string in nth harmonic is given by

$$f_n = n.\frac{v}{2l} \qquad \text{...(i)}$$

where, v = speed of sound and l = length of string.



$\frac{3V}{2\ell} = 240$

$\frac{V}{2\ell} = 80$ (fundamental frequency)

$\frac{V}{2 \times 2} = 80$

$V = 320$ m/sec (velocity)

16. The position of a particle as a function of time t, is given by

$x(t) = at + bt^2 - ct^3$

where a, b and c constants. When the particle attains zero acceleration, then its velocity will be:

A. $a + \frac{b^2}{4c}$

B. $a + \frac{b^2}{c}$

C. $a + \frac{b^2}{2c}$

D. $a + \frac{b^2}{3c}$

Ans. D

Sol:

Given that,

$x = at + bt^2 - ct^3$

$v = a + 2bt - 3\,ct^2$

acceleration $= 2b - 6ct = 0$

$f = \frac{2b}{6c} = \frac{b}{3c}$

so velocity at,

$\Rightarrow t = \frac{b}{3c}$

$$v = a + 2b\frac{b}{3c} - 3c\frac{b^2}{9c^2}$$
$$v = \frac{3ac + 2b^2}{3c} - \frac{3cb^2}{9c^2}$$
$$v = \frac{3ac + 2b^2}{3c} - \frac{\cancel{3}\cancel{c}b^2}{\cancel{9}\cancel{c^2}}$$
$$v = \frac{3ac + 2b^2}{3c} - \frac{b^2}{3c}$$
$$v = \frac{3ac + 2b^2 - b^2}{3c}$$
$$v = \frac{3ac + b^2}{3c}$$
$$v = \frac{3ac}{3c} + \frac{b^2}{3c}$$
$$v = \frac{\cancel{3}a\cancel{c}}{\cancel{3c}} + \frac{b^2}{3c}$$
$$v = a + \frac{b^2}{3c}$$

17. Moment of inertial of a body about a given axis is 1.5 kg m^2. Initially the body is at rest. In order to produce a rotational kinetic energy of 1200 J, the angular acceleration of 20 rad/s^2 must be applied about the axis for a duration of

A. 2 s | B. 5 s
C. 2.5 s | D. 3 s

Ans. A

Sol:

Rotation kinetic energy of a body is given by

$$KE_{rotational} = \frac{1}{2} I\omega^2$$

$\omega_1 = 0, \alpha = 20$
$\omega_2 = \omega_1 + \alpha t$
$\omega_2 = 20t$
$\frac{1}{2} I\omega^2 = 1200$
$\frac{1}{2} 1.5 \times 400\ t^2 = 1200$
$t = 2\,sec$

18. A He^+ ion is in its first excited state. Its ionization energy is:

A. 6.04 eV B. 54.40 eV

C. 13.60 eV D. 48.36 eV

Ans. C

Sol:

$E = -13.6 \frac{z^2}{n^2} eV$
$Z = 2,$
$n = 2$
$E = -13.6\ eV$
ionisation energy is
$\Rightarrow E = -13.6 \frac{z^2}{n^2} eV$
$\Rightarrow E = -13.6 \frac{2^2}{2^2} eV$
$= +13.6\ eV$

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19. Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm. coming from a distant object, the limit of resolution of the telescope is close to:

A. 2.0×10^{-7} rad B. 4.5×10^{-7} rad

C. 1.5×10^{-7} rad D. 3.0×10^{-7} rad

Ans. D

Sol:

Rayleigh's criterion is the range of resolution for the telescope,

Given that,

$\lambda = 600 \times 10^{-9}$

$d = 250 \times 10^{-2}$

We know that,

$$\text{resolution limit}^{+} = \frac{1.22\lambda}{d}$$

$$= \frac{1.22 \times 600 \times 10^{-9}}{250 \times 10^{-2}}$$

$$= \frac{1.22 \times 600 \times 10^{-9} \times 10^{2}}{250}$$

$= 2.9 \times 10^{-7}$ rad

$= 3.0 \times 10^{-7}$

20. The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A. What resistance must be connected to it in order to convert it into an ammeter of range 0 – 0.5 A?

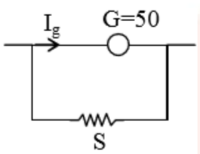
A. 0.5 ohm B. 0.2 ohm

C. 0.02 ohm D. 0.002 ohm

Ans. B

Sol:

Ammeter circuit is shown in the figure below



So, $I_g G = (I - I_g)S$

Here, $I_g = 0.002$ A,

I = 0.5A,

G = 50 Ω

$$I_y = 0.002 \text{ A}$$
$$2(0.5 - 0.002) = 50 \times 0.002$$
$$2 = \frac{0.1}{0.498} = 0.2$$

21. The physical sizes of the transmitter and receiver antenna in a communication system are:

A. inversely proportional to carrier frequency
B. inversely proportional to modulation frequency
C. independent of both carrier and modulation frequency
D. proportional to carrier frequency

Ans. A

Sol:

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Size of antenna is directly proportional to the wavelength of the signal.

Also, the speed at which signal moves = carrier frequency × wavelength

$\Rightarrow \quad f\lambda = c \Rightarrow \lambda \propto \frac{1}{f}$

∴ size of antenna $\propto \frac{1}{f}$.

22. A metal wire of resistance 3 Ω is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle 60° at the centre, the equivalent resistance between these two points will be:

A. $\frac{12}{5}\Omega$ B. $\frac{5}{3}\Omega$

C. $\frac{5}{2}\Omega$ D. $\frac{7}{2}\Omega$

Ans. C

Sol:

Initial resistance of wire is 3 Ω. Let its length is l and area is A.

Then, $R_{initial} = \rho\frac{l}{A} = 3\Omega$

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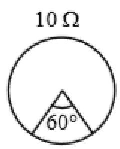
When length become double

$R = 12\,\Omega, \left(R = \frac{\rho\ell}{A} = \frac{\rho\ell^2}{V}\right)$



$R \propto \ell$

$R_{Eff} = \frac{10}{6}$

$= \frac{5}{3}$

23. A thin smooth rod of length L and mass M is rotating freely with angular speed ω_0 about an axis perpendicular to the rod and passing through its center. Two beads of mass m and negligible size are at the center of the rod initially. The beads are free to slide along the rod. The angular speed of the system, when the beads reach the opposite ends of the rod, will be:

A. $\frac{M\,\omega_0}{M+6m}$

B. $\frac{M\,\omega_0}{M+2m}$

C. $\frac{M\,\omega_0}{M+m}$

D. $\frac{M\,\omega_0}{M+3m}$

Ans. A

Sol:

As there is no external torque on system.

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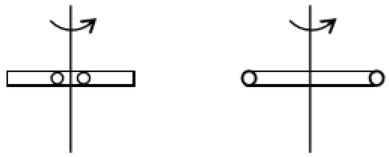
$\therefore$ Angular momentum of system is conserved.

$\Rightarrow I_i\omega_i = I_f\omega_f$



$J_1 = J_2$ (Angular momentum conservation)

$$\frac{M\ell^2}{12}\omega_0 = \left(\frac{M\ell^2}{12} + \frac{2m\ell^2}{4}\right)\omega$$

$$\omega = \frac{M\omega_0}{M + 6m}$$

24. A convex lens of local length 20 cm produces images of the same magnification 2 when an object is kept at two distance x_1 and x_2 ($x_1 > x_2$) from the lens. The ratio of x_1 and x_2 is:

A. 3 : 1 B. 4 : 3

C. 2 : 1 D. 5 : 3

Ans. A

Sol:

Magnification for a lens can also be written as

$$m = \left(\frac{f}{f + u}\right)$$

when m = -2 (for real image)

$$-2 = \frac{f}{f + x_1}$$

$$\Rightarrow -2f + (-2)x_1 = f \text{ or } x_1 = -\frac{3f}{2}$$

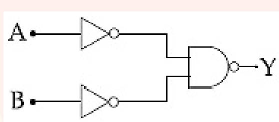
Similarly, when m = +2 (for virtual image)

$$+2 = \frac{f}{f + x_2} \Rightarrow 2f + 2x_2 = f \text{ or } x_2 = \frac{-f}{2}$$

Now, the ratio of x_1 and x_2 is

$$\frac{x_1}{x_2} = \frac{-\frac{3f}{2}}{-\frac{f}{2}} = \frac{3}{2}$$

25. The logic gate equivalent to the given logic circuit is:



A. NAND
B. AND
C. NOR
D. OR

Ans. D

Sol:

The given logic gate circuit can be drawn as shown below

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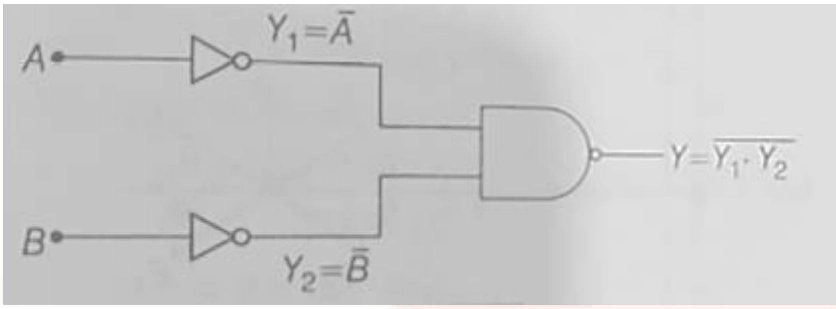

Here, $Y = \overline{Y_1.Y_2} = \overline{\overline{A}.\overline{B}}.$

Using de-Morgan's theorem, i.e.,

$\overline{x.y} = \overline{x} + \overline{y}$

$\therefore\ Y = \overline{\overline{A}} + \overline{\overline{B}} = A + B \qquad \left[\because \overline{\overline{x}} = x\right]$

This represents the Boolean expression for OR gate.

26. The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\,\hat{i} + \left(4 = 20t^2\right)\hat{j}$. What is the magnitude of the acceleration at t = 1?

A. 40 B. 100

C. 50 D. 25

Ans. C

Sol:

Position vector of particle is given as

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$\vec{r}(t) = 15t^2\,\hat{i} + \left(4 - 20t^2\right)\hat{j}$

Velocity of particle is

$$v = \frac{dr}{dt} = \frac{d}{dt} \qquad \left[15t^2\hat{i} + \left(4 - 20t^2\right)\hat{j}\right]$$
$$= 30t\,\hat{i} - 40t\,\hat{j}$$

acceleration $= 30i - 40j$

$$a = \sqrt{30^2 + (-40)^2}$$
$$= 50$$

27. The particle of mass 'm' is moving with speed '2υ' and collides with a mass '2m' moving with speed 'υ' in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass 'm', which move at angle 45° with respect to the original direction.

The speed of each of the moving particle will be:

A. $\sqrt{2}\,\upsilon$ B. $\upsilon / \sqrt{2}$

C. $2\sqrt{2}\,\upsilon$ D. $\upsilon / \left(2\sqrt{2}\right)$

Ans. C

Sol:

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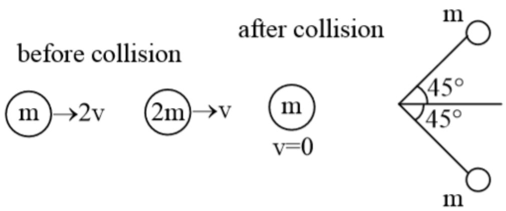

As we know that in collision, linear momentum is conserved in both x and y direction separately.

So, $(p_x)_{initial} = (p_x)_{final}$

Momentum conservation in x directon $2mv + 2mv = mv'\cos 45° + mv'\cos 45°$

$$4v = \sqrt{2}\, v'$$
$$v' = 2\sqrt{2}\, v$$

28. A moving coil galvanometer has a coil with 175 turns and area 1 cm^2. It uses torsion band of torsion constant 10^{-6} N-m/rad. The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA. The value of B (in Tesla) is approximately.

A. 10^{-3} B. 10^{-4}

C. 10^{-2} D. 10^{-1}

Ans. A

Sol:

In a moving coil galvanometer in equilibrium, torque on coil due to current is balanced by torque of torsion band.

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As, torque on coil,

$\tau = M \times B = NIAB \sin \alpha$

Where, B = magnetic field strength

I = current

N = number of turns of coil.

$\tau = \vec{M} \times \vec{B}$

$C\dot{\theta} = NIAB$

$10^{-6} \dfrac{\pi}{180} = 10^{-3} \times 10^{-4} \times 175\,B$

$B = 10^{-3}\ T$

29. The area of a square is 5.29 cm^2. The area of 7 such squares taking into account the significant figures is:

A. 37.030 cm^2 B. 37.03 cm^2

C. 37 cm^2 D. 37.0 cm^2

Ans. B

Sol:

Area of 1 square = 5.29 cm^2

Area of seven such squares

= 7 times addition of area of 1 square

= 5.29 + 5.29 + 5.29 … 7 times

= 37.03 cm^2

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30. A massless spring (k = 800 N/m), attached with a mass (500 g) is completely immersed in 1 kg of water. The spring is stretched by 2cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations shop completely?

(Assume that the water container and spring receive negligible heat and specific heat mass = 400 J/kg K, specific heat of water = 4184 J/kg K)

A. 10^{-5} K B. 10^{-1} K

C. 10^{-4} K D. 10^{-3} K

Ans. A

Sol:

When vibrations of mass are suddenly stopped, oscillation energy (or stored energy of spring) is dissipated as heat, causing rise of temperature. So, conversation of energy gives

$$\frac{1}{2}Kx_m^2 = (m_1\ S_2 + m_2 s_2)\Delta T$$

$$\frac{1}{2}800\left(\frac{2}{100}\right)^2 = (0.5\times 400 + 1\times 4184)\ \Delta T$$

$$\frac{1600}{4264\times 10^4} = \Delta T$$

$$\Delta T = 3\times 10^{-5}$$

Order 10^{-5} K

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Chemistry

1. During compression of a spring the work done is 10 kJ and 2 kJ escaped to the surroundings as heat. The change in internal energy, ΔU (in kJ) is:

A. –8 B. 12

C. –12 D. 8

Ans. D

Sol:

In the given system, during the compression of a spring the work done is 10 kJ and 2 kJ of heat is escaped to the surroundings.

So, q = -2kJ and W = 10 kJ

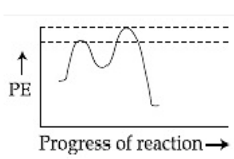
Work done on system = +10 kT

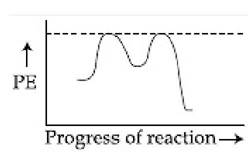
Heat escaped = –2kI

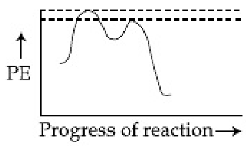
$\Delta U = q + w$

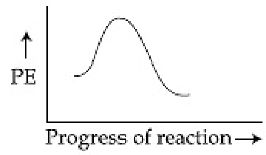
$= 10 - 2 = 8$ KJ

2. Which of the following potential energy (PE) diagrams represents the S_N1 reaction?

A. 

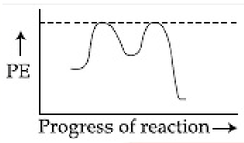
B. 

C. 

D. 

Ans. B

Sol:



3. The structures of beryllium chloride in the solid state and vapour phase, respectively, are:

A. chain and dimeric
B. dimeric and dimeric
C. chain and chain
D. dimeric and chain

Ans. A

Sol:

The structure of beryllium chloride in the solid state and vapour phase, respectively are dimeric and chain. In vopur phase at above 900°C, $BeCl_2$ is monomeric having a linear structure Cl – Be – Cl.

Solid state-chain

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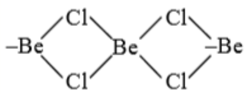
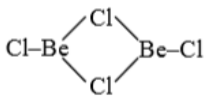
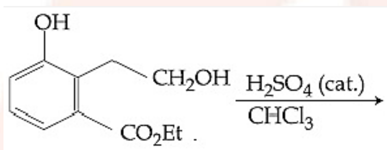
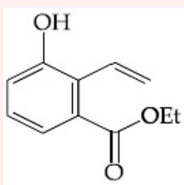


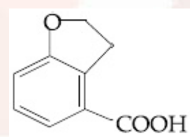
In vapour stae dimeric

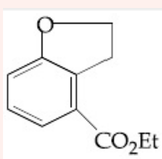


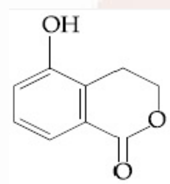
4. The major product of the following reaction is:



A. 

B. 

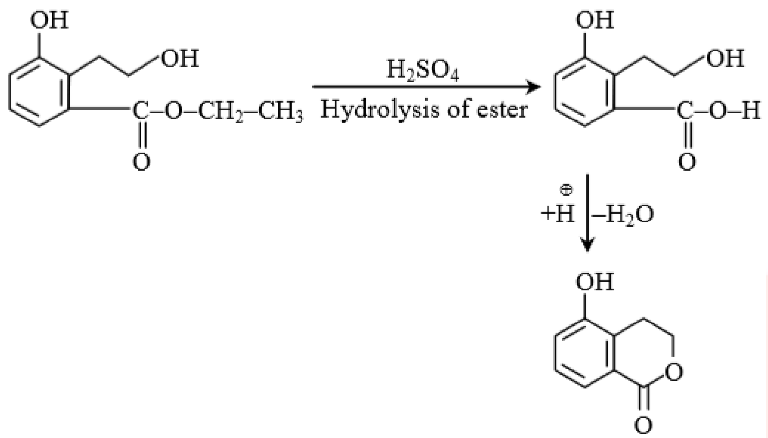
C. 

D. 

Ans. D

Sol:

Given reaction involves acidic hydrolysis of esters followed by the intramolecular cryclisation. The chemical equation is as follows:



5. The one that is not a carbonate ore is:

A. calamine B. bauxite

C. malachite D. siderite

Ans. B

Sol:

Malachite = $CuCO_3. Cu(OH)_2$

Bauxite = $Al_2O_3. 2H_2O$ or $AlO_x .(OH)_3^{-2x}$ (where $0 < x < 1$)

Calamine = $ZnCO_3$ Siderite = $FeCO_3$

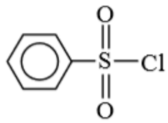
6. Hinsberg's reagent is:

A. C_6H_5COCl B. $(COCl)_2$

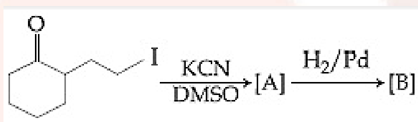
C. $C_6H_5SO_2Cl$ D. $SOCl_2$

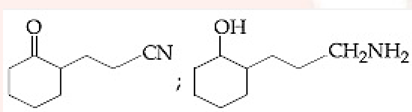
Ans. C

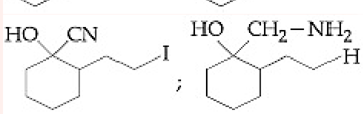
Sol: Hinsberg's reagent is benzene sulphonyl chloride or $C_6H_5SO_2C\ell$

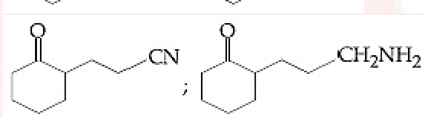


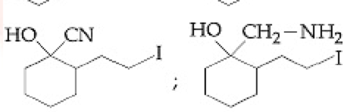
7. The major products A and B for the following reactions are, respectively:



A. 

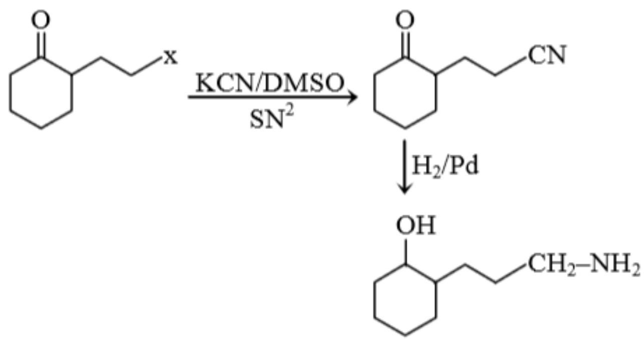
B. 

C. 

D. 

Ans. B

Sol:



8. Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect? (The Bohr radius is represented by a_0).

A. The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.
B. The probability density of finding the electron is maximum at the nucleus.
C. The electron can be found at a distance $2a_0$ from the nucleus.
D. The magnitude of the potential energy is double that of its kinetic energy on an average.

Ans. A

Sol: The total energy of the electron is minimum when it is at a distance a0 from the nucleus for 1 s orbital

9. 10 mL of 1 mM surfactant solution forms a monolayer covering 0.24 cm^2 on a polar substrate. If the polar head is approximated as a cube, what is its edge length?

A. 0.1 nm B. 1.0 pm

C. 2.0 nm D. 2.0 pm

Ans. D

Sol:

Given, volume = 10 mL

Molarity = 1 mM = 10^{-3} M

$$\text{Moles} = \frac{mv_{ml}}{1000} = \frac{10^{-3}\times 10}{1000} = 10^{-5}\text{ mole}$$

$10^{-5}\ N_A$ molecules covering area $= 0.24\ cm^2$

$1\ N_A$ molecules covering area $= \dfrac{0.24}{10^{-5}\ N_A}\ cm^2$

$$= \frac{0.24}{10^{-5}\times 6\times 10^{23}} = a^2$$

$$a^2 = 4\times 10^{-20}\ cm^2$$

$$a = 2\times 10^{-10} cm$$

$$a = 2\times 10^{-12}\ m$$

$$a = 2\ pm$$

10. In the following reaction carbonyl compound $+ \text{MeOH} \xrightleftharpoons{\text{HCl}} \text{acetal}$ Rate of the reaction is the highest for:

A. Propanal as substrate and methanol is stoichiometric amount

B. Acetone as substrate and methanol in excess

C. Propanal as substrate and methanol in excess

D. Acetone as substrate and methanol in stoichiometric amount

Ans. C

Sol:

Generally, aldehydes are more reactive than ketones in nucleophilic addition reactions.

$\therefore$ Rate of reaction with alcohol to form acetal. Only aldehydes are responsible for the function of acetals. Propanal as substrate & methanol is in excess

11. Molal depression constant for a solvent is 4.0 K kg mol^{-1}. The depression in the freezing point of the solvent for 0.03 mol kg^{-1} solution of K_2SO_4 is:

(Assume complete dissociation of the electrolyte)

A. 0.12 K   B. 0.18 K

C. 0.36 K   D. 0.24 K

Ans. C

Sol:

Depression in freezing point (ΔT_f) is

Given by $\Delta T_f = iK_f m$

i = vant Hoff factor

K_f = molal depression constant

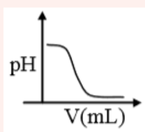
m = molality

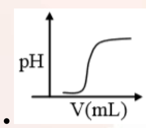
$\Delta T_f = ik_f m$

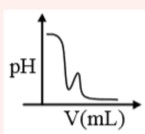
= 3(4) (0.03)

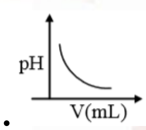
$\Delta T_f = 0.36$

12. In an acid-base titration, 0.1 M HCl solution was added to the NaOH solution of unknown strength. Which of the following correctly shows the change of pH of the titration mixture in this experiment?

1. 

2. 

3. 

4. 

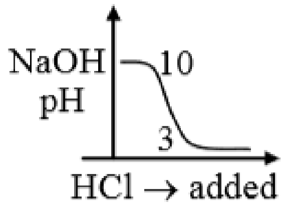
A. 1 B. 2

C. 3 D. 4

Ans. C

Sol:

The graph that shows the correct change of pH of the titration mixture in the experiment.



13. The maximum number of possible oxidation states of actinoides are shown by:

A. neptunium (Np) and plutonium (Pu)
B. actinium (Ac) and thorium (Th)
C. berkelium (Bk) and californium (Cf)
D. nobelium (No) and lawrencium (Lr)

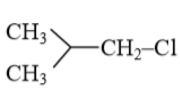
Ans. A

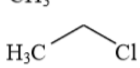
Sol:

Ac	Th	P	U	Np	Pu	An	Cm	Bk	Cf	Es	Fm	Md
3		3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3
	4	+4	+4	+4	+4	+4	4	4				
		5	+5	+5	+5	+5						
			6	+6	+6	+6						
				7	7							

$\therefore$ Maximum oxidation state is shown by (Np and Pu)

14. Increasing order of reactivity of the following compounds for S_N1 substitution is:

A. 

B. 

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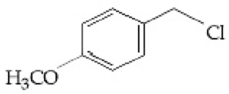
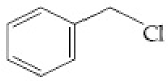
C. 

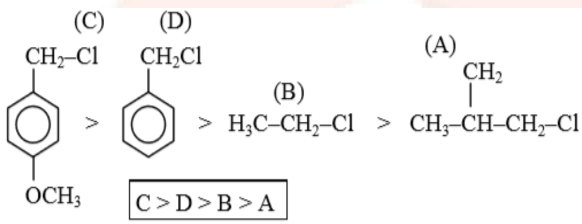
D. 

A. (A) < (B) < (D) < (C) B. (B) < (C) < (D) < (A)

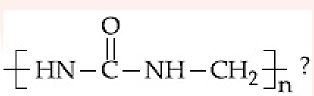
C. (A) < (B) < (D) < (C) D. (B) < (C) < (A) < (D)

Ans. A

Sol:



15. Which of the following compounds is a constituent of the polymer

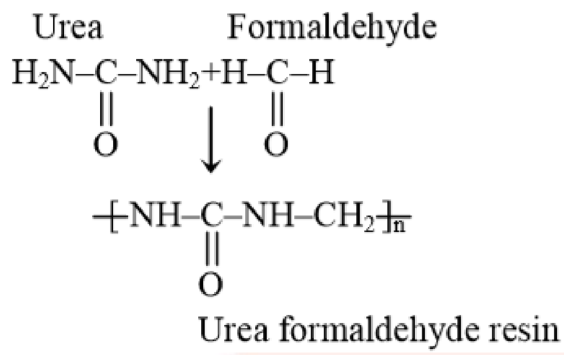


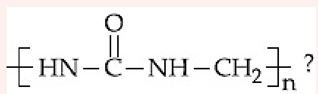
A. Ammonia B. Methylamine

C. Formaldehyde D. N-Methyl urea

Ans. C

Sol:



So, from the above reaction we have Formaldehyde is a constituent of the polymer .

16. The layer of atmosphere between 10 km above the sea level is called as:

A. stratosphere B. thermosphere

C. troposphere D. mesosphere

Ans. A

Sol:

The layer of atmosphere between 10 km above the sea level is called as stratosphere.

17. The correct statements among I to III are:

I. Valence bond theory cannot explain the color exhibited by transition metal complexes.

II. Valence bond theory can predict quantitatively the magnetic properties of transition metal complexes.

III. Valence bond theory cannot distinguish ligands as weak and strong field ones.

A. I, II and III B. I and II only

C. II and III only D. I and III only

Ans. D

Sol:

Among the given statements, correct statements are I and III only.

18. A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?

A. 0.10 B. 0.20

C. 0.15 D. 0.05

Ans. D

Sol:

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0.1 F of electricity is passed through Ni(NO3)2 solution

$\therefore$ Amount of Ni deposited = 0.1 eq

$\therefore$ Moles $= \frac{0.1}{2} = 0.05$

19. Assertion:

For the extraction of iron, haematite ore is used.

Reason:

Haematite is a carbonate ore of iron.

A. Only the assertion is correct.
B. Both the assertion and reason are correct and the reason is the correct explanation for the assertion.
C. Only the reason is correct.
D. Both the assertion and reason are correct, but the reason is not the correct explanation for the assertion.

Ans. A

Sol:

Only assertion is correct and reason is incorrect. Haematite is not a carbonate ore.

Extration fo Fe is done from Haematite or this is true but reason is strong as Haematitte is Fe_2O_3.

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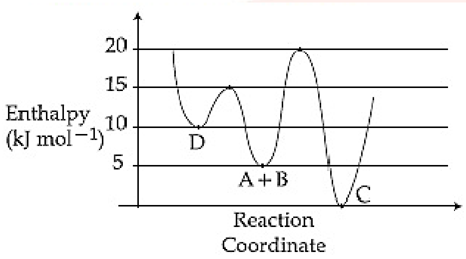
20. Consider the given plot of enthalpy of the following reaction between A and B.

$A + B \rightarrow C + D.$

Identify the incorrect statement.



A. D is kinetically stable product.
B. Formation of A and B from C has highest enthalpy of activation.
C. Activation enthalpy to form C is 5 kJ mol^{-1} less than that to form D.
D. C is the thermodynamically stable product.

Ans. C

Sol:

Only statement (d) is incorrect. Corrected statement is "Activation enthalpy to form C is 15 kg mol^{-1} more than 5 kg mol^{-1} that is required to form D."

$E_a = (d \rightarrow c) = 15 - 0 = 15$

$E_a = (A + B) \rightarrow C = 15$

$E_a = (A + B \rightarrow D) = 10$

$E_a = C \rightarrow (A + B) = 20$

21. HF has highest boiling point among hydrogen halides, because it has:

A. lowest dissociation enthalpy
B. strongest van der Waal’s interactions
C. lowest ionic character
D. strongest hydrogen bonding

Ans. D

Sol:

HF has highest boiling point among hydrogen halides because it has strongest hydrogen bonding.

22. The correct statements among I to III regarding group 13 element oxides are,

I. Boron trioxide is acidic.

II. Oxides of aluminium and gallium are amphoteric.

III. Oxides of Indium and thallium are basic.

A. I and II only   B. II and III only

C. I and III only   D. I, II and III

Ans. A

Sol:

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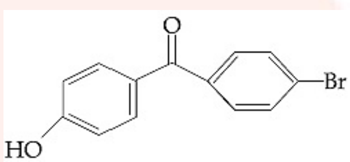
(1) VBT does not explain colour exhibited by complex of transition metal.

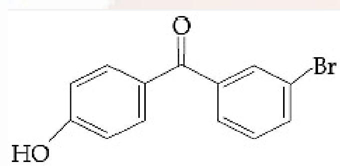
(2) VBT does not distinguish between strong and weak complex.

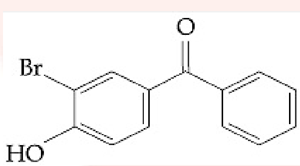
(3) VBT does not give quantitative interpretation of magnetic properties.

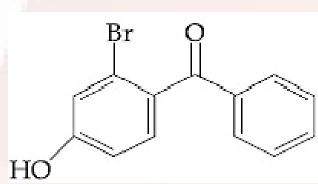
∴ Ascending to question statement I and III are correct.

23. p-Hydroxybenzophenone upon reaction with bromine in carbon tetrachloride gives:

A. 

B. 

C. 

D. 

Ans. C

Sol:

p-hydroxy benzophenone upon reaction with bromine in carbon tetrachloride gives 3-bromo-4-hydroxy benzophenone.

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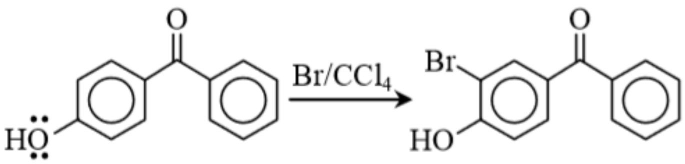


(M effect)

+M effect having group is responsible for activation towards ESR

24. Noradrenaline is a/an:

A. Antidepressant
B. Antihistamine
C. Neurotransmitter
D. Antacid

Ans. C

Sol: Nonadrenaline is a neurotransmitter.

25. The peptide that gives positive ceric ammonium nitrate and carbylamines tests is:

A. Gln – Asp   B. Lys - Asp

C. Asp - Gln   D. Ser – Lys

Ans. D

Sol:

Ceric ammonium nitrate test is given by alcohol. Only serine(ser) contain –OH group.

26. At a given temperature T, gases Ne, Ar, Xe and Kr are found to deviate from ideal gas behaviour. Their equation of state is given as $p = \frac{RT}{V - b}$ at T.

Here, b is the van der Waals constant. Which gas will exhibit steepest increase in the plot of Z (compression factor) vs p?

A. Ne B. Ar

C. Xe D. Kr

Ans. C

Sol:

Noble gases such as Ne, Ar, Xe and Kr found to deviate from ideal gas behavior.

The given equation is

$$P = \frac{RT}{(V-b)}$$

$$p(V-b) = RT$$

$$\left(P + \frac{a}{V^2}\right)(V-b) = RT$$

At high pressure

$$P(V-b) = RT$$

$$PV - Pb = RT$$

$$\frac{PV}{RT} - \frac{Pb}{RT} = 1$$

$$Z = 1 + \frac{Pb}{RT}$$

$$Z > 1, Z \propto b$$

27. What would be the molality of 20% (mass/mass) aqueous solution of KI? (molar mass of KI = 166 g mol^{-1})

A. 1.08 B. 1.51

C. 1.35 D. 1.48

Ans. B

Sol:

20% W/W KI solution

i.e. 100 g solution contains 20 g KI

$\therefore$ Mass of solvent = 100 – 20 = 80 g

$$\text{Molality} = \frac{\text{gm (solute)}}{\text{mv} \times \text{kg(sovlent)}}$$

$$= \frac{20 \times 1000}{166 \times 80}$$

$$= 1.506$$

$$= 1.51$$

28. Among the following species, the diamagnetic molecule is:

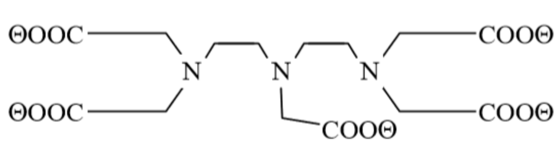
A. B_2 B. O_2

C. NO D. CO

Ans. D

Sol: CO is diamagnetic in nature.

29. The maximum possible denticities of a ligand given below towards a common transition and inner-transition metal ion, respectively, are:



A. 8 and 6 B. 8 and 8

C. 6 and 8 D. 6 and 6

Ans. C

Sol: General C.No. of CN^- in transition elements is 6 and in inner transition elements is 8 – 12.

30. The amorphous form of silica is:

A. cristobalite B. Kieselguhr

C. quartz D. tridymite

Ans. B

Sol: Kieselguhr is amorphous form of silica.

Mathematics

1. The vertices B and C of a ΔABC lie on the line, $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that BC = 5 units. Then the area (in sq. units) of this triangle, given that the point A(1, –1, 2), is:

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A. $5\sqrt{17}$ B. 6

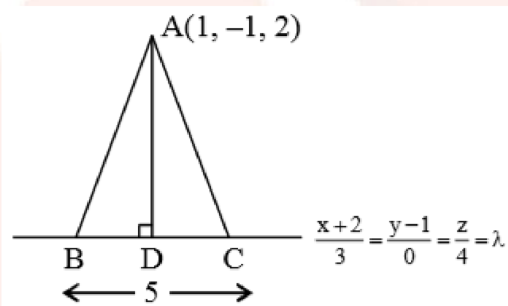
C. $2\sqrt{34}$ D. $\sqrt{34}$

Ans. D

Sol:

Given line is,

$$\frac{x+2}{3}=\frac{y-1}{0}=\frac{z}{4}=\lambda\ (\text{let})\ldots(i)$$



Since, point D lies on the line BC.

Let any point on given line is

$D(3\lambda-2, 1, 4\lambda)$

Now $AD \perp BC$

D.R. of BC $\Rightarrow a_1=3, b_1=0, c_1=4$

D.R. of AD $\Rightarrow a_2=3\lambda-3, b_2=2, c_2=4\lambda-2$

We know that,

$\Rightarrow a_1a_2+b_1b_2+c_1c_2=0$

$\Rightarrow 3(3\lambda-3)+0\times 2+4(4\lambda-2)=0$

$\Rightarrow 9\lambda-9+0+16\lambda-8=0$

$\Rightarrow 25\lambda=17$

$\Rightarrow \lambda=\frac{17}{25}$

co-oridnation of point $D\left(\frac{1}{25}, 1, \frac{68}{25}\right)$

$$AD = \sqrt{\frac{576}{625} + 4 + \frac{324}{25}}$$

$$AD = \sqrt{\frac{576}{625} + \frac{100+324}{25}} = \sqrt{\frac{576}{625} + \frac{424}{25}}$$

$$AD = \sqrt{\frac{576+10625}{625}}$$

$$= \frac{2}{5}\sqrt{34}$$

Area of $\Delta ABC = \frac{1}{2} \times BC \times AD$

$$= \frac{1}{2} \times 5 \times \frac{2}{5}\sqrt{34}$$

$$= \sqrt{34}$$

2. If the sum and product of the first three terms in an A.P. are 33 and 1155, respectively, then a value of its 11th term is:

A. –36 B. –25

C. –35 D. 25

Ans. B

Sol:

Explain that there are three numbers in A.P.

a – d, a, a + d

Given that ,

a – d + a + d = 33

$a - \cancel{d} + a + \cancel{d} = 33$

$2a = 33$

$\Rightarrow a = 11$

and (a-d)(a)(a+d) = 1155

$\Rightarrow a(a^2 - d^2) = 1155$

$\therefore a = 11$

$\Rightarrow 11((11^2) - d^2) = 1155$

$\Rightarrow 11(121 - d^2) = 1155$

$\Rightarrow d^2 = 16$

$\Rightarrow d = \pm 4$

If d = 4

then, first term $a - d = 7$

If d = -4

then, first term $a - d = 15$

Now,

$T_{11} = 7 + 10(4) = 7 + 40$

$T_{11} = 47$

and

$T_{11} = 15 + 10(-4) = 15 + (-40)$

$T_{11} = -25$

3. The sum of the series 1 + 2 × 3 + 3 × 5 + 4 × 7 + …. upto 11th term is:

A. 916 B. 946

C. 915 D. 945

Ans. B

Sol:

S = 1 + 2 × 3 + 3 × 5 + 4 × 7 + ...+ upto 11th terms

nth term of the series is,

We know that,

$T_n = n(2n-1)$

$\Rightarrow \; S = \sum_{n=1}^{11} Tn = \sum_{n=1}^{11}\left(2n^2 - n\right)$

$\Rightarrow \; S_n = \frac{2n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$

put the value of $n = 11$

$\Rightarrow \; S_{11} = \frac{2(11)-(11+1)(2\times 11+1)}{6} - \frac{11\times(11+1)}{2}$

$\Rightarrow \; S_{11} = \frac{2(11)(12)(23)}{6} - \frac{11(12)}{2}$

$\Rightarrow \; S_{11} = \frac{2\times 11\times 12\times 13 - 3\times 11\times 12}{6}$

$\Rightarrow \; S_{11} = \frac{22\times 12\times 13 - 33\times 12}{6}$

$\Rightarrow \; S_{11} = 946$

4. The area (in sq. units) of the region $A = \left\{(x, y) : \frac{y^2}{2} \le x \le y + 5\right\}$ is:

A. 18 B. 16

C. 30 D. $\frac{53}{3}$

Ans. A

Sol:

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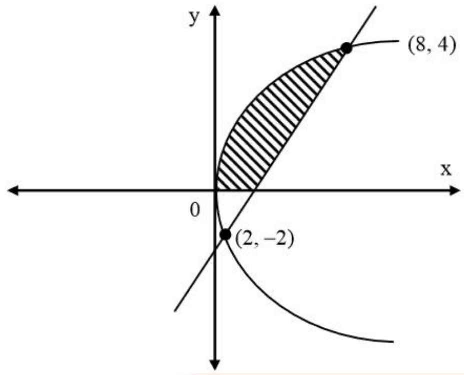

$y^2 = 2x$........(i)
and $x - y - 4 = 0$........(ii)
solving this equations (i) and (ii)
$(x-4)^2 = 2x$
$\Rightarrow x^2 - 10x + 16 = 0$
$\Rightarrow x = 8_1 + 2$ and $y = 4_1 - 2$

$$A = \int_{-2}^{4} \left(y + 4 - \frac{y^2}{2} \right) dy$$

$$A = \left(\frac{y^2}{4} + 4y - \frac{y^3}{3} \right)_{-2}^{4}$$

Taken this limit,

$$A = \left(4 + 16 - \frac{64}{3} \right) - \left(1 - 8 + \frac{8}{6} \right)$$

$$A = \left(\frac{3 \times 20 - 64}{3} \right) - \left(\frac{6 \times (-7) + 8}{6} \right)$$

$A = 18$

5. If $\cos x \frac{dy}{dx} - y \sin x = 6x, \left(0 < x < \frac{\pi}{2} \right)$ and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

A. $-\frac{\pi^2}{2}$

B. $-\frac{\pi^2}{2\sqrt{3}}$

C. $-\frac{\pi^2}{4\sqrt{3}}$ D. $\frac{\pi^2}{2\sqrt{3}}$

Ans. B

Sol: Given that ,

$$\cos x\frac{dy}{dx} - y\sin x = 6x$$

$$\Rightarrow \frac{dy}{dx} - y\tan x = 6x\sec x$$

$$\text{I.F} = e^{-\int \tan x\, dx} = e^{-\log_e \sec x} = \frac{1}{\sec x}$$

$\therefore$ Solution of equation

$$\Rightarrow y.\frac{1}{\sec x} = \int 6x\sec x.\frac{1}{\sec x}dx$$

$$\Rightarrow \frac{y}{\sec x} = 3x^2 + c \qquad ....(i)$$

given,

$$y\left(\frac{\pi}{3}\right) = 0$$

So,

$$0 = \frac{3\pi^2}{9} + C$$

$$\Rightarrow C = -\frac{\cancel{3}\pi^2}{\cancel{9}}$$

$$\Rightarrow \ C = -\frac{\pi^2}{3}$$

Now, from equation (1)

$$\Rightarrow \ \frac{y}{\sec x} = 3x^2 - \frac{\pi^2}{3}$$

$$\text{at}, x = \frac{\pi}{6}$$

$$\Rightarrow \ \frac{y}{\sec\frac{\pi}{6}} = 3(\frac{\pi}{6})^2 - \frac{\pi^2}{3}$$

$$\Rightarrow \frac{\sqrt{3}y}{2} = \frac{3\pi^2}{36} - \frac{\pi^2}{3}$$

$$\Rightarrow \ y = -\frac{\pi^2}{2\sqrt{3}}$$

6. If some three consecutive coefficients in the binomial expansion of $(x+1)^n$ in powers of *x* are in the ratio 2 : 15 : 70, then the average of these three coefficients is:

A. 232 B. 625

C. 964 D. 227

Ans. A

Sol:

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Given binomial is $(x + 1)^n$, whose general term, is $T_{r+1} = {}^nC_rx^r$

$$\text{given}: \frac{{}^nC_{r-1}}{{}^nC_r} = \frac{2}{15} \Rightarrow \frac{r}{n-r+1} = \frac{2}{15}$$

$\Rightarrow\ 15r = 2n - 2r + 2$

$\Rightarrow 15r + 2r = 2n + 2$

$\Rightarrow\ 17r = 2n + 2 \quad ....(i)$

also given,

$$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{15}{17}$$

$$\Rightarrow\ \frac{r+1}{n-r} = \frac{3}{14}$$

$\Rightarrow\ 3n - 3r = 14r + 14$

$\Rightarrow\ 3n - 14 = 14r + 3r$

$\Rightarrow\ 17r = 3n - 14.....(ii)$

Solving this equations (i) and (ii)

$n = 16, r = 2$

$$\text{average of coefficinet} = \frac{{}^{16}C_1 + {}^{16}C_2 + {}^{16}C_3}{3}$$

$$= \frac{16 + 120 + 560}{3}$$

$= 232$

7. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z), then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to:

A. $-\frac{1}{4}$

B. $\frac{3}{4}$

C. -4

D. $\frac{1}{2}$

Ans. D

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Sol:

Given system of linear equations

$2x + 3y - z = 0$

$x + ky - 2z = 0$

and $2x - y + z = 0$.

Given system of equations has non-trival solution.

$$\therefore \ \Delta \Longrightarrow \begin{vmatrix} 2 & 3 & -1 \\ 1 & K & -2 \\ 2 & -1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow K = \frac{9}{2}$$

so equation are

$2x + 3y - z = 0$ ....(1)

$x + 9/2\ y - 2z = 0$ ....(2)

$2x - y + z = 0$ ....(3)

$(1) - (3) \Longrightarrow 4y - 2z = 0$

$\Longrightarrow 2y = z$ ....(4)

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$\Rightarrow \boxed{\frac{y}{z} = \frac{1}{2}}$

From equation (1) and (4)

$2x + 3y - 2y = 0$

$\Rightarrow 2x + y = 0$

$\Rightarrow \boxed{\frac{x}{y} = \frac{-1}{2}}$

or $\boxed{\frac{z}{x} = -4}$

$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + K = \frac{1}{2}$

8. If $f : R \to R$ is differentiable function and $f(2) = 6$, then $\lim_{x \to 2} \int_{6}^{f(x)} \frac{2t\ dt}{(x-2)}$ is:

A. $12f'(2)$ B. 0

C. $24f'(2)$ D. $2f'(2)$

Ans. A

Sol:

Use formula

$$\frac{d}{dx}\int_{\phi_1(x)}^{\phi_2(x)} f(t)\,dt = f\left[\phi_2(x)\right]\cdot\phi'_2(x) - f\left[\phi_1(x)\right].\phi'_1(x)$$

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$\lim_{x\to 2} \int_6^{f(x)} \frac{2t\,dt}{(x-2)} dx$ $\{\text{given that } f(2) = 6\}$

$\frac{0}{0}$ form. So we use L-Hospital Rule

$= \lim_{x\to 2} \frac{f'(x).2f(x)}{1}$

$= f'(2) \,.\, 2f(2)$

$= 12f'(2)$

9. The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point:

A. (4, –2) B. (6, –2)

C. (–6, 4) D. (–4, 6)

Ans. B

Sol:

Given circles are

$x^2 + y^2 = 4$, centre $c_1(0, 0)$ and radius $r_1 = 2$ and $x^2 + y^2 + 6x + 8y - 24 = 0$, centre c_2 (-3, -4) and radius $r_2 = 7$.

$\Longrightarrow d = c_1c_2 = 5$

also $d = |r_1 - r_2|$

circles touch externally

equation of common tangent $s_1 - s_2 = 0$ $\Longrightarrow 6x + 8y - 20 = 0$

$\Longrightarrow 3x + 4y - 10 = 0$

Point (6, –2) satisfy it

10. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10 m; is:

A. 1/15π　　B. 1/10π

C. 1/5π　　D. 2/π

Ans. C

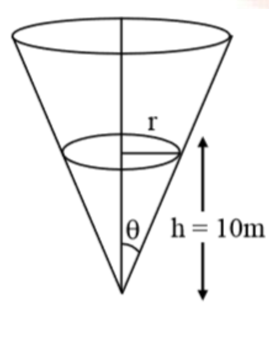
Sol:

We know that,

Volume of cone $\frac{1}{3}\pi r^2 h$,

Where r = radius and h = height of the cone.

It is given that $\theta = \tan^{-1}\left(\frac{1}{2}\right)$



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Now,

$$\Rightarrow \tan\theta = \frac{1}{2} = \frac{r}{h}$$

$$\Rightarrow r = \frac{h}{2}$$

Since, $v = \frac{1}{3}\pi\frac{h^3}{4}$

So, differentiating both sides, we get,

$$\frac{dv}{dt} = \frac{\pi}{12}\left(3h^2\right)\frac{dh}{dt}$$

$$S = \frac{\pi}{4}(100)\frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{1}{5\pi}$$

Thus, the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10 m; is 1/5π.

11. Let $z \in C$ be such that $|z| < 1$. If $\omega = \frac{5+3z}{5(1-z)}$

A. 5 Im(ω) < 1 B. 5 Re(ω) > 1

C. 4 Im(ω) > 5 D. 5 Re(ω) > 4

Ans. B

Sol: It is given that

$$\omega = \frac{5+32}{5(1-Z)}$$

$$\Rightarrow \; 5\omega - 5\omega z = 5 + 3z$$

$$\Rightarrow \; z = \frac{5\omega - 5}{3 + 5\omega}$$

And, it is also given that $|z| < 1$,

Therefore,

$$\Rightarrow \; \left|\frac{5\omega - 5}{3 + 5\omega}\right| < 1$$

$$\Rightarrow \; |5\omega - 5| < |3 + 5\omega|$$

$$\Rightarrow \; |\omega - 1| < \left|\frac{3}{4} + \omega\right|$$

12. If $p \Rightarrow (q \vee r)$ is false, then the truth values of p, q, r are respectively:

A. T, F, F
B. F, T, T
C. T, T, F
D. F, F, F

Ans. A

Sol: Given that

$p \Rightarrow \; (q \vee r)$ is false

$(\because \; T \; \Rightarrow F = F)$

So, $P = T, q = F$ and $r = F$

13. The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}, (x, y \in R, x \neq y)$ for which $A^T A = 3I_3$ is:

A. 2 B. 4

C. 3 D. 6

Ans. B

Sol: Given that

Given matrix

$$A = \begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix}, (x, y \in R, x \neq y)$$

for which

$$(A^T)(A) = 3I_3$$

$$\Rightarrow \begin{bmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 8x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow 8x^2 = 3 \Rightarrow x = \pm\sqrt{\frac{3}{8}}$$

$$\Rightarrow 6y^2 = 3 \Rightarrow y = \pm\sqrt{\frac{1}{2}}$$

4 materices are possible

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14. If the tangent to the parabola $y^2 = x$ at a point (α, β), $(\beta > 0)$ is also a tangent to the ellipse, $x^2 + 2y^2 = 1$, then α is equal to:

A. $\sqrt{2}+1$ B. $2\sqrt{2}+1$

C. $2\sqrt{2}-1$ D. $\sqrt{2}-1$

Ans. A

Sol:

Since the point (α, β) is on the parabola $y^2 = x$, so $\alpha = \beta^2$

Now, equation of tangent at point (α, β) to the parabola $y^2 = x$, is T = 0

at (α, β) is $T = 0$

$$y\beta = \frac{x+\alpha}{2}$$

$$\Rightarrow \; y\beta = \frac{x+\beta^2}{2} \qquad (\therefore \; \beta^2 = \alpha)$$

$$\Rightarrow \; y = \frac{1}{2\beta}x + \frac{\beta}{2}$$

$$\left(m = \frac{2}{2\beta}, c = \frac{\beta}{2}\right)$$

this is also a tangent to ellipre $x^2 + 2y^2 = 1$

$\therefore\ C = \pm\sqrt{a^2 m^2 + b^2}$

$\Rightarrow\ \frac{\beta}{2} = \pm\sqrt{\frac{1}{4\beta^2} + \frac{1}{2}}$

$\Rightarrow\ \frac{\beta^2}{4} = \frac{1}{4\beta^2} + \frac{1}{2}$

$\Rightarrow\ \beta^4 - 2\beta^2 - 1 = 0$

$\Rightarrow\ (\beta - 1)^2 = 2$

$\Rightarrow\ \beta^2 - 1 = \sqrt{2} \quad (\because\ \beta > 0)$

$\Rightarrow\ \beta^2 = \sqrt{2} + 1$

$\alpha = \beta^2 = \sqrt{2} + 1$

15. If $f(x) = [x] - \left[\frac{x}{4}\right], x \in R$, where [*x*] denotes the greatest integer function, then:

A. *f* is continuous at *x* = 4.
B. $\lim_{x\to4-} f(x)$ exists but $\lim_{x\to4+} f(x)$ does not exist.
C. Both $\lim_{x\to4-} f(x)$ and $\lim_{x\to4+} f(x)$ exist but are not equal.
D. $\lim_{x\to4+} f(x)$ exists but $\lim_{x\to4-} f(x)$ does not exist.

Ans. A

Sol: Given that

$f(x) = [x] - \left[\frac{x}{4}\right]$

$\lim_{x\to4^+} f(x) = \lim_{x\to4^+}\left\{[x] - \left[\frac{x}{4}\right]\right\} = 4 - 1 = 3$

$\lim_{x\to4^-} f(x) = \lim_{x\to4^-}\left\{[x] - \left[\frac{x}{4}\right]\right\} = 3 - 0 = 3$

$f(4) = 3$

$\because \lim_{x \to 4^-} f(x) = f(4) = \lim_{x \to 4^+} f(x) = 3$

So, the functions f(x) is continuous at x = 4.

16. The area (in sq. units) of the smaller of two circles that touch the parabola, $y^2 = 4x$ at the point (1, 2) and the x-axis is:

A. $4\pi\left(3+\sqrt{2}\right)$ B. $4\pi\left(2-\sqrt{2}\right)$

C. $8\pi\left(2-\sqrt{2}\right)$ D. $8\pi\left(3-2\sqrt{2}\right)$

Ans. D

Sol:

Given region $A = \left\{(x,\ y) : \frac{y^2}{2} \le x \le y + 4\right\}$

$\therefore \frac{y^2}{2} = x$

$\Rightarrow y^2 = 2x$ …(i)

and x = y + 4 ⇒ y = x – 4 ...(ii)

Graphical representation as below:

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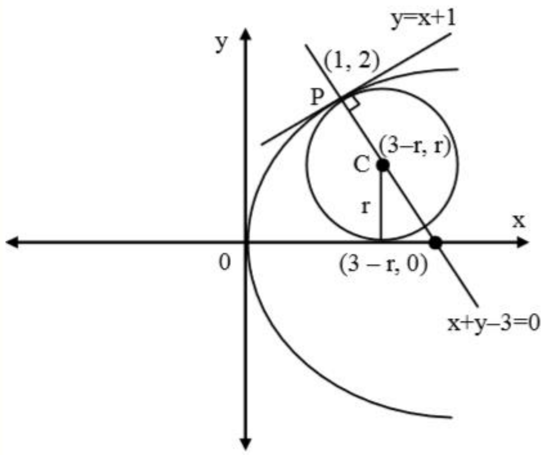

equation of tangent to the perabola $y^2 = 4x$ at (1, 2) is

$$2y = 4\left(\frac{x+1}{2}\right)$$
$$\Rightarrow \ y = x + 1$$

equation of normal $y = -x + 3$

Let center be $c(3 - r, r)$

Now $PC^2 = r^2$

$\Longrightarrow (3 - r - 1)^2 + (r - 2)^2 = r^2$

$\Longrightarrow 2(2 - r)^2 = r^2$

$\Longrightarrow r^2 - 8r + 8 = 0$

$\Rightarrow \ r = 4 \pm 2\sqrt{2}$

for $r = 4 + 2\sqrt{2}$ $\quad (3 - r < 0)$ Not possible

So, $r = 4 - 2\sqrt{2}$

Area $= \pi r^2 = \pi\left(16 + 8 - 16\sqrt{2}\right)$

$= 8\pi\left(3 - 2\sqrt{2}\right)$

17. The domain of the definition of the function $f(x)=\frac{1}{4-x^2}+\log_{10}\left(x^3-x\right)$ is:

A. $(-1,0)\cup(1,2)\cup(2,\infty)$
B. $(1,2)\cup(2,\infty)$
C. $(-1,0)\cup(1,2)\cup(3,\infty)$
D. $(-2,-1)\cup(-1,0)\cup(2,\infty)$

Ans. A

Sol: It is given that

$$f(x)=\frac{1}{4-x^2}+\log_{10}\left(x^3-x\right)$$

Let,

$$f_1=\frac{1}{4-x^2}$$
$$\Rightarrow\ 4-x^2\neq 0$$
$$\Rightarrow\ x\neq \pm 2$$

And,

$$f_2=\log_{10}\left(x^3-x\right)$$
$$x^3-x>0$$
$$\Rightarrow\ x(x-1)(x+1)>0$$

Now, this implies that the value of f_1 from -1 to 1and value of f_2 is from 1 to infinity.

Thus,

$$x\in(-1,0)\cup(1,\infty)-\{2\}$$
$$x\in(-1,0)\cup(1,2)\cup\{2,\infty\}$$

18. If

$\int e^{\sec x}(\sec x \tan x\, f(x) + (\sec x \tan x + \sec^2 x))dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is:

A. $x\sec x + \tan x + \frac{1}{2}$

B. $\sec x - \tan x - \frac{1}{2}$

C. $\sec x + \tan x + \frac{1}{2}$

D. $\sec x + x\tan x - \frac{1}{2}$

Ans. C

Sol: It is given that

$$\int e^{\sec x}\left(\sec x \tan x\, f(x) + \left(\sec x \tan x + \sec^2 x\right)\right)dx$$
$$= e^{\sec x} f(x) + C$$

Now, differentiating both sides, we get,

$$\Rightarrow \quad e^{\sec x}\left(\sec x \tan x f(x) + \left(\sec x \tan x + \sec^2 x\right)\right)$$
$$= e^{\sec x}.\sec x \tan x\, f(x) + e^{\sec x} f'(x)$$
$$\Rightarrow \quad f'(x) = \sec^2 x + \tan x \sec x$$
$$\Rightarrow \quad f(x) = \tan x + \sec x + C$$

Thus, a possible choice of $f(x)$ is $\sec x + \tan x + \frac{1}{2}$.

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19. The value of sin10° sin30° sin50° sin70° is:

A. $\frac{1}{18}$ B. $\frac{1}{36}$

C. $\frac{1}{32}$ D. $\frac{1}{16}$

Ans. C

Sol: It is given that

sin10° sin30° sin50° sin70°

= sin30°{sin10° sin(60°–10°) sin(60° + 10°)}

$= \sin 30° \left\{\frac{1}{4}\sin 3(10)°\right\}$

$= \frac{1}{2}\left(\frac{1}{4} \times \frac{1}{2}\right)$

$= \frac{1}{16}$

Thus, the value of sin10° sin30° sin50° sin70° is $\frac{1}{16}$.

20. Let P be the plane, which contains the line of intersection of the planes, $x + y + x - 6 = 0$ and $2x + 3y + z + 5 = 0$ and it is perpendicular to the xy-plane. Then the distance of the point (0, 0, 256) from P is equal to:

A. $63\sqrt{5}$ B. $205\sqrt{5}$

C. $11/\sqrt{5}$ D. $17/\sqrt{5}$

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Ans. C

Sol:

Equation of plane $P_1 + \lambda P_2 = 0$

$(x + y + z - 6) + \lambda(2x + 3y + z + 5) = 0$

$\Rightarrow x(1 + 2\lambda) + y(1 + 3\lambda) + z(1 + \lambda) - 6 + 5\lambda = 0$

$\therefore 1 + \lambda = 0$

$\Rightarrow \lambda = -1$

So, equation of plane

$-x - 2y - 11 = 0$

$\Rightarrow x + 2y + 11 = 0$

distance of the point (0, 0, 256) from this plane.

$$= \left|\frac{0 + 0 + 1}{\sqrt{1 + 4}}\right| = \frac{11}{\sqrt{5}}$$

21. If a unit vector $\vec{a}$ makes angles $\pi/3$ with $\hat{i}$, $\pi/4$ with $\hat{j}$ and $\theta \in (0, \pi)$ with k, then a value of θ is:

A. $\frac{5\pi}{6}$

B. $\frac{2\pi}{3}$

C. $\frac{\pi}{4}$

D. $\frac{5\pi}{12}$

Ans. B

Sol:

Given unit vector a makes and angel $\pi/3$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and $\theta \in (0, \pi)$ with $\hat{k}$.

As we know that

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2\gamma = 1$

$\Rightarrow \cos^2\gamma = 1 - \frac{3}{4} = \frac{1}{4}$

$\Rightarrow \cos\gamma = \pm\frac{1}{2}$

$\Rightarrow \gamma = \frac{\pi}{3}$ or $\frac{2\pi}{3}$

22. The mean and the median of the following ten numbers in increasing order10, 22, 26, 29, 34, *x*, 42, 67, 70, *y* are 42 and 35 respectively, then $\frac{y}{x}$ is equal to:

A. 9/4 B. 7/3

C. 7/2 D. 8/3

Ans. B

Sol: It is given that mean is 42.

So, we have,

$\Rightarrow \frac{10+22+26+29+34+x+42+67+70+y}{10} = 45$

$\Rightarrow 420 = 300 + x + y$

$\Rightarrow x + y = 120 \quad \ldots(i)$

And, It is also given that median is 35.

Hence,

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$$\Rightarrow \frac{34+x}{2} = 35$$
$$\Rightarrow x = 36$$

Now, putting the value of x is equation (1), we get,

$y = 84$

Thus, $\frac{y}{x} = \frac{84}{36} = \frac{7}{3}$

23. Two newspapers A and B are published in a city. It is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisements is:

A. 13.9 B. 13.5

C. 13 D. 12.8

Ans. A

Sol:

Let population = 100

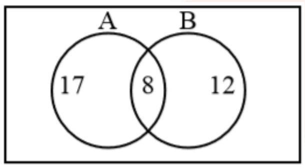
$n(A) = 25$

$n(B) = 20$

$n(A \cap B) = 8$

$n(A \cap \bar{B}) = 17$

$n(\bar{A} \cap B) = 12$



According to the question, Percentage of the population who look into advertisement is

$$= \left[\frac{30}{100} \times n\left(A \cap \bar{B}\right)\right] + \left[\frac{40}{100} \times n\left(\bar{A} \cap B\right)\right] + \left[\frac{50}{100} \times n(A \cap B)\right]$$

$$= \left(\frac{30}{100} \times 17\right) + \left(\frac{40}{100} \times 12\right) + \left(\frac{50}{100} \times 8\right)$$

= 5.1 + 4.8 + 4 = 13.9

24. The value of the integral

$\int_0^1 x \cot^{-1}\left(1 - x^2 + x^4\right) dx$ is:

A. $\frac{\pi}{4} - \frac{1}{2}\log_e 2$ B. $\frac{\pi}{4} - \log_e 2$

C. $\frac{\pi}{2} - \frac{1}{2}\log_e 2$ D. $\frac{\pi}{2} - \log_e 2$

Ans. A

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Sol: Given that

$$I=\int_0^1 x\cot^{-1}\left(1-x^2+x^4\right)dx$$

$$I=\int_0^1 x\tan^{-1}\left(\frac{1}{1-x^2+x^4}\right)dx$$

$$I=\int_0^1 x\tan^{-1}\left\{\frac{x^2-\left(x^2-1\right)}{1+x^2\left(x^2-1\right)}\right\}dx$$

$$I=\int_0^1 x\left\{\tan^{-1}x^2-\tan^{-1}\left(x^2-1\right)\right\}dx$$

Let $x^2=t \Rightarrow 2x\,dx=dt$

$$I=\frac{1}{2}\int_0^1\left\{\tan^{-1}t-\tan^{-1}(t-1)\right\}dt$$

$$I=\frac{1}{2}\int_0^1\tan^{-1}t\,dt-\frac{1}{2}\int_0^1\tan^{-1}(t-1)\,dt$$

$$=\frac{1}{2}\int_0^1\tan^{-1}t\,dt-\frac{1}{2}\int_0^1\tan^{-1}(-t)\,dt$$

$$=\frac{1}{2}\int_0^1\tan^{-1}t\,dt-\frac{1}{2}\int_0^1\tan^{-1}(t)\,dt$$

$$I=\int_0^1\tan^{-1}(t)\,dt$$

$$=\left(\tan.\tan^{-1}\right)_0^1-\int_0^1\frac{t}{1+t^2}dt$$

$$=\left(\frac{\pi}{4}\right)-\frac{1}{2}\left[\log\left(1+t^2\right)\right]_0^1$$

$$=\frac{\pi}{4}-\frac{1}{2}\log_e 2$$

25. Some identical balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row consists of two balls and so on. If 99 more identical balls are added to the total number of balls used in forming the equilateral triangle, then all these balls can be arranged in a square whose each side contains exactly 2 balls less than the number of balls each side of the triangle contains. Then the number of balls used to form the equilateral triangle is:

A. 262 B. 190

C. 157 D. 225

Ans. B

Sol: Balls used in equilateral triangle= $\frac{n(n+1)}{2}$

Here side of equilateral triangle has n-balls

No. of balls in each side of square is = (n – 2)

Given that

$$\frac{n(n+1)}{2} + 99 = (n-2)^2$$
$$\Rightarrow n^2 + n + 198 = 2n^2 - 8n + 8$$
$$\Rightarrow n^2 - 9n - 190 = 0$$
$$\Rightarrow (n-19)(n+10) = 0$$
$$\Rightarrow n = 19$$
$$\text{number of balls} = \frac{19.20}{2} = 190$$

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26. Two poles standing on a horizontal ground are of heights 5 m and 10 m respectively. The line joining their tops makes an angle of 15° with the ground. Then the distance (in m) between the poles, is:

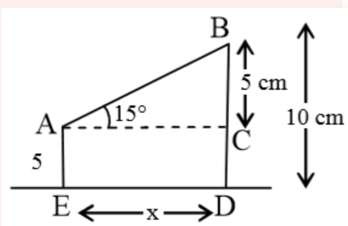
A. $\frac{5}{2}\left(2+\sqrt{3}\right)$ B. $10\left(\sqrt{3}-1\right)$

C. $5\left(\sqrt{3}+1\right)$ D. $5\left(2+\sqrt{3}\right)$

Ans. D

Sol:

Given heights of two poles are 5m and 10 m



i.e., from figure BD = 10 cm, AE = 5 cm

$$\text{in } \Delta ABC \Rightarrow \tan 15^\circ = \frac{5}{x}$$

$$\Rightarrow 2-\sqrt{3} = \frac{5}{x}$$

$$\Rightarrow x = 5\left(2+\sqrt{3}\right)$$

27. If the two lines $x + (a - 1)\, y = 1$ and $2x + a^2 y = 1$ ($a \in R - \{0, 1\}$) are perpendicular, then the distance of their point of intersection from the origin is:

A. $\frac{2}{5}$ B. $\frac{\sqrt{2}}{5}$

C. $\sqrt{\frac{2}{5}}$ D. $\frac{2}{\sqrt{5}}$

Ans. C

Sol:

If lines are perpendicular to each other, then product of their slopes is -1, i.e., $m_1m_2 = -1$

As we know that

Two lines are perpendicular

$\therefore\ m_1m_2 = -1$

$\Rightarrow\ \left(\frac{-1}{a-1}\right)\left(\frac{-2}{a^2}\right) = -1$

$\Rightarrow\ a^3 - a^2 + 2 = 0$

$\Rightarrow\ (a+1)(a^2 - 2a + 2) = 0$

$\therefore\ a = -1$

so, lines are $L_1 : x - 2y + 1 = 0$

$L_2 : 2x + y - 1 = 0$

Solving there equation we get point of intersetion $P\left(\frac{1}{5}, \frac{3}{5}\right)$

Distance between two points (x_1, y_1) and

$(x_2 y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Now distance of P from origin

$OP = \sqrt{\frac{1}{25} + \frac{9}{25}} = \sqrt{\frac{2}{5}}$

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28. If the function $f(x)=\begin{cases} a|\pi - x| + 1, & x \le 5 \\ b|x-\pi| + 3, & x > 5 \end{cases}$ is continuous at $x = 5$, then the value of a – b is:

A. $\frac{2}{\pi - 5}$ B. $\frac{2}{\pi + 5}$

C. $\frac{2}{5 - \pi}$ D. $\frac{-2}{\pi + 5}$

Ans. C

Sol: Given that

$$f(x)=\begin{cases} a\,|\,\pi - x\,| + 1, & x < 5 \\ b\,|\,\pi - x\,| + 3, & x > 5 \end{cases}$$

continous at $x = 5$

$\therefore$ L.H.L = R.H.L = f(5)

$\Rightarrow\; b\,|\,\pi - 5\,| + 3 = a\,|\,\pi - 5\,| + 1$

$\Rightarrow\; -b(\pi - 5) + 3 = -a(5 - \pi) + 1$

$\Rightarrow\; (a - b)(\pi - 5) = -2$

$\Rightarrow\; a - b = \frac{-2}{\pi - 5} = \frac{2}{5 - \pi}$

29. A rectangle is inscribed in a circle with a diameter lying along the line $3y = x + 7$. If the two adjacent vertices of the rectangle are (–8, 5) and (6, 5), then the area of the rectangle (in sq. units) is:

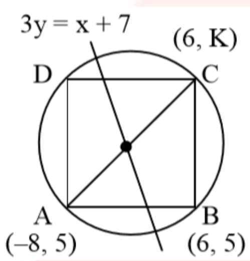
A. 72 B. 98

C. 56 D. 84

Ans. D

Sol:

Given points are (-8, 5) and (6, 5) in which y-coordinate is same, i.e, these points lie on horizontal lien y = 5.



Let vertex c is (6, K) then center of circle $\left(-1, \frac{5+K}{2}\right)$ it lies on diameter 3y = x + 7.

$\Rightarrow 3\left(\frac{5+K}{2}\right) = -1 + 7$

$\Rightarrow \; K = -1$

So AB = 14 and BC = 6

Area = 14 × 6 = 84

30. If m is chosen in the quadratic equation $(m^2 + 1)\, x^2 - 3x\, (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is:

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A. $8\sqrt{3}$ B. $4\sqrt{3}$

C. $8\sqrt{5}$ D. $10\sqrt{5}$

Ans. C

Sol:

Given quadratic equation is

$(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$

Let the roots of quadratic Eq. (i) are α and β, so

$\alpha + \beta = \frac{3}{m^2 + 1}$ and $\alpha\beta = m^2 + 1$

According to the question, the sum of roots is greatest and it is possible only when "$(m^2 + 1)$ is minimum" and minim value of $m^2 + 1 = 1$,

when m = 0

$$\left(m^2 + 1\right)x^2 - 3x + \left(m + 1\right)^2 = 0$$

$$\Rightarrow \quad \alpha + \beta = \frac{3}{m^2 + 1}$$

$$\alpha\beta = \frac{\left(m + 1\right)^2}{m^2 + 1}$$

$\because \quad \alpha + \beta$ is maximum

$\therefore \quad m^2 + 1$ is maximum

$\Rightarrow \quad m = 0$

$\therefore \quad \alpha + \beta = 3$ and $\alpha\beta = 1$

$\left|\alpha^3 - \beta^3\right| = \left|(\alpha - \beta)\left(\alpha^2 + \alpha\beta + \beta^2\right)\right|$

$= \left|\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}\ \left\{(\alpha + \beta)^2 - \alpha\beta\right\}\right|$

$= \left|\sqrt{9 - 4}\,(9 - 1)\right|$

$= 8\sqrt{5}$

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